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CO-2

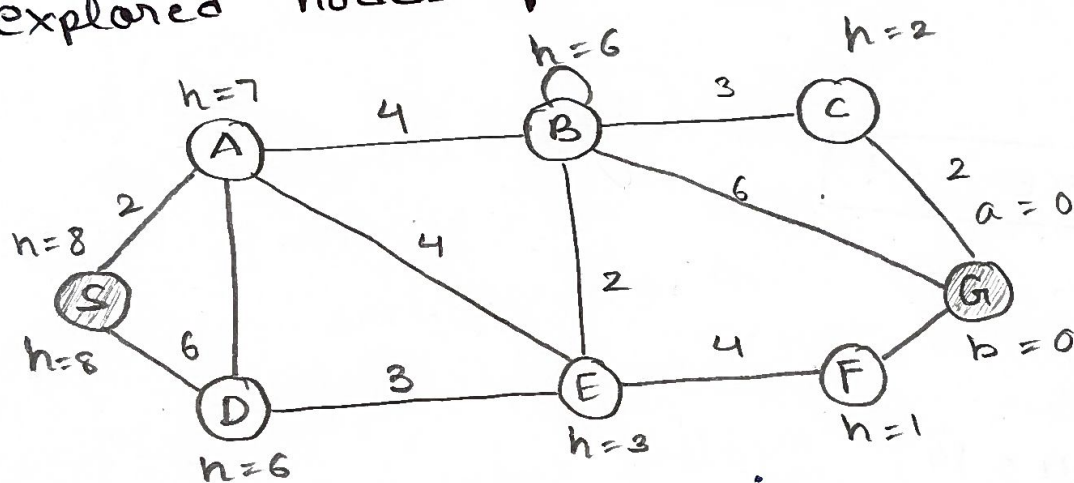
10/8/26

ASSESSMENT TOOL - 2

★ PROBLEM SOLVING ASSIGNMENT ★

i) Smart Ambulance Navigation:

Use A* search to find the optimal route from the hospital to the emergency location, showing the explored nodes, $f(n)$ values and route cost.



A* evaluation function is:

$$f(n) = g(m) + h(n)$$

Where:

$g(m)$ = actual cost from starting node S.

$h(n)$ = heuristic / estimated cost from current node to goal G

$f(n)$ = evaluation value

given:

$$h(S) = 8$$

$$h(A) = 7$$

$$h(B) = 6$$

$$h(C) = 2$$

$$h(D) = 6$$

$$h(E) = 3$$

$$h(F) = 1$$

$$h(G) = 0$$

① Step - 1:

$$S \rightarrow A$$

$$\Rightarrow f(A) = g(A) + h(A) \Rightarrow f(A) = 2 + 7 = 9$$

$$\therefore \boxed{S \rightarrow A = 9}$$

② Step - 2:

$$S \rightarrow A \rightarrow B$$

$$\Rightarrow g(B) = 2 + 4 = 6$$

$$f(B) = 6 + 6 = 12$$

$$\Rightarrow S \rightarrow A \rightarrow D$$

$$g(D) = 2 + 6 = 8$$

$$f(D) = 8 + 6 = 14$$

$$S \rightarrow A \rightarrow E$$

$$\Rightarrow g(E) = 2 + 4 = 6$$

$$f(E) = 6 + 3$$

$$f(E) = 9$$

$$\therefore \boxed{S \rightarrow A \rightarrow E = 9}$$

③ Step - 3:

$$S \rightarrow A \rightarrow E \rightarrow F$$

$$g(F) = 2 + 4 + 4 = 10$$

$$h(F) = 1$$

$$f(F) = 10 + 1 = 11$$

$$\therefore \boxed{S \rightarrow A \rightarrow E \rightarrow F = 11}$$

$$S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$$

④ Step - 4:

$$\boxed{S \rightarrow A \rightarrow E \rightarrow F \rightarrow G} \leftarrow \text{final path}$$

$\therefore$ The final path is $S \rightarrow A \rightarrow E \rightarrow F \rightarrow G$ using A* search algorithm.

Autonomous Warehouse Robot:

Design an online search strategy for a robot that discovers obstacles while moving and show how it replans when an obstacle is found.

5	S	?	?	?	?	?	?
4	?	#	?	?	#	?	?
3	?	?	?	#	?	?	#
2	?	#	?	?	?	#	?
1	?	?	#	?	?	?	G
	1	2	3	4	5	6	7

S - charging station (start)

G - Target shelf (Goal)

- Obstacle (Unknown Initially)

? - Unknown cell

The robot starts at S and must reach G. Since all obstacles are not known initially, online A* search is suitable. The robot whenever it discovers an obstacle.

Algorithm

$$f(n) = g(m) + h(n)$$

where $g(m)$ is the path cost and $h(n)$ is the estimated distance to G.

Execution

Initially, the robot follows the best available path:
 $S \rightarrow (2,5) \rightarrow (3,5) \rightarrow (4,5) \rightarrow (5,5) \rightarrow (6,5) \rightarrow (7,5)$

Then:

$(7,5) \rightarrow (7,4)$

At $(7,4)$ it discovers an obstacle at $(7,3)$.

$(7,4)$

$\times (7,3) \leftarrow$ obstacle

↑
replan

The robot updates its map and chooses another path:

$(7,4) \rightarrow (6,4) \rightarrow (6,3)$

It then discovers another obstacle at $(6,2)$ and replans again.

$(6,3) \rightarrow (5,3) \rightarrow (5,2) \rightarrow (5,1) \rightarrow (6,1) \rightarrow G$

Final path:

$S \rightarrow (2,5) \rightarrow (3,5) \rightarrow (4,5) \rightarrow (5,5) \rightarrow (6,5) \rightarrow (7,5) \rightarrow (7,4) \rightarrow (7,3) \rightarrow (7,2) \rightarrow (7,1) \rightarrow (6,1) \rightarrow (6,2) \rightarrow (6,3) \rightarrow (5,3) \rightarrow (5,2) \rightarrow (5,1) \rightarrow (6,1)$

Online vs offline

<u>Online Search</u>	<u>Offline Search</u>
obstacles discovered during movement	obstacles known initially
Replans when environment changes	Plans once using complete map
suitable for the problem	Not suitable

In conclusion: Online A* allow the robot to observe $\rightarrow$ update $\rightarrow$ replan $\rightarrow$ move, enabling it to reach G despite initially unknown obstacles.

3) University Examination scheduling:

Formulate the exam timetable as a CSP and solve it using MRV, Forward checking and Backtracking satisfying conflict and hall - capacity constraints.

Examination Hall	Capacity
Hall A	120 students
Hall B	90 students
Hall C	60 students

Examination	E1	E2	E3	E4	E5	E6
No. of students	100	80	70	60	90	50

Variables :

$\{E_1, E_2, E_3, E_4, E_5, E_6\}$

Domain :

$\{T_1, T_2, T_3, T_4\}$

Constraints :

$E_1 \neq E_2, E_1 \neq E_3$

$E_2 \neq E_4, E_3 \neq E_5$

$E_4 \neq E_6, E_2 \neq E_5$

MVRV + Forward checking

1. $E_2 = T_1 \rightarrow$ remove T_1 from E_1, E_4, E_5

2. $E_1 = T_2 \rightarrow$ remove T_2 from E_3

3. $E_4 = T_2 \rightarrow$ remove T_2 from E_6

4. $E_5 = T_2$

5. $E_3 = T_1$

6. $E_6 = T_1$

Final Timetable

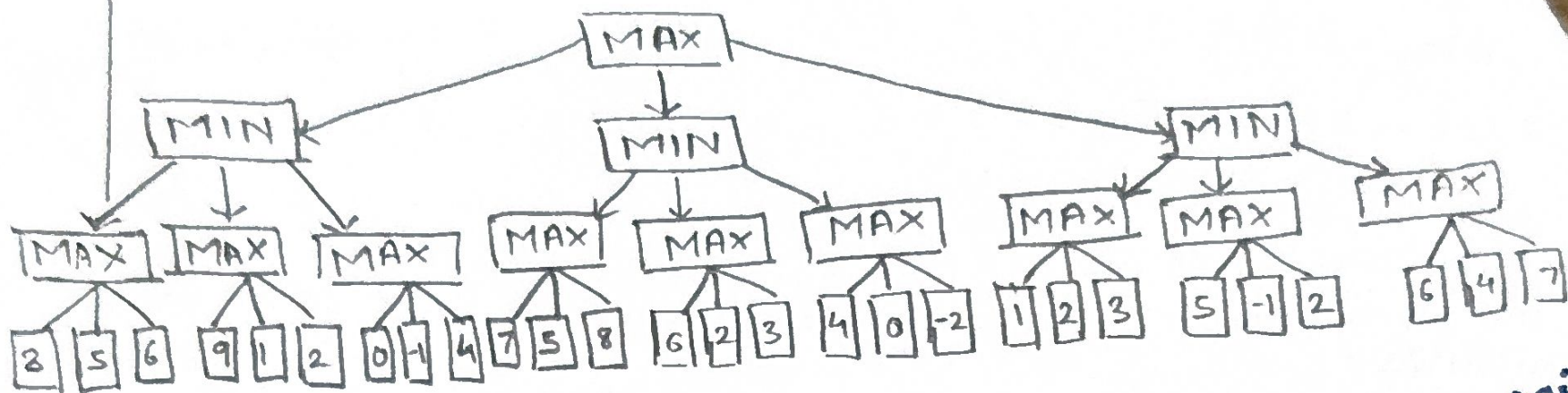
<u>slot</u>	<u>Hall A</u>	<u>Hall B</u>	<u>Hall C</u>
T_1	$E_3(70)$	$E_2(80)$	$E_6(50)$ ✓
T_2	$E_1(100)$	$E_5(90)$	$E_4(60)$

All students conflicts are satisfied and all examinations fit within the given hall capacities.

$\therefore$ Final timetable is valid and conflict-free. ✓

4) Game - Playing AI :

Apply Minimax and Alpha - Beta Pruning to the given game tree, determine the best move and identify the pruned branches.



The given game tree is evaluated using minimax.
The AI is the MAX player, while the opponent is MIN.

★ Minimax calculation

For the three main branches:

$$\text{Left} = 4$$

$$\text{Middle} = 4$$

$$\text{Right} = 3$$

Therefore, at the MAX root:

$$\text{MAX}(4, 4, 3) = 4$$

Hence, the AI should choose:

Left or Middle move

★ Alpha-Beta Pruning

After obtaining $\alpha = 4$ at the root, the Right branch produces:

$$\beta = 3$$

since:

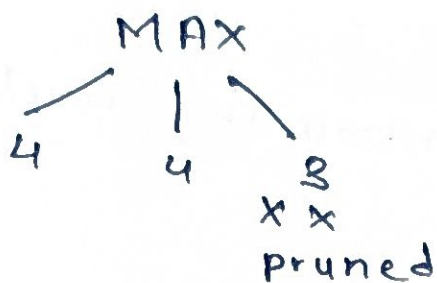
$$\beta \leq \alpha$$

the remaining branches under the Right node do not need to be evaluated.

Also, in the left subtree, after obtaining a value of 9 while $\beta = 6$,

$$\alpha \geq \beta$$

so the remaining values 1 & 2 are pruned.



∴ Minimax value = 4 //

Best move = Left or Middle //

Alpha-Beta pruning reduces unnecessary node evaluation while producing the same Minimax result.

5) Delivery Route Optimization:

Apply Hill climbing to improve the delivery route, determine the final solution, discuss whether it is globally optimal and explain how to escape a local optimum.

Distance Matrix (km)

	D	A	B	C	E
D	-	6	9	8	7
A	6	-	3	5	6
B	9	3	-	4	5
C	8	5	4	-	3
E	7	6	5	3	-

Hill climbing is a local search optimization algorithm used to minimize the total delivery distance. The assignment asks to apply it to the given initial route and neighbouring routes.

① Step-1: start

initial route: SS

cost = 23 km

② Step-2: check Neighbours

The best neighbouring route has: 22 km

since:

$$22 < 23$$

Hill climbing moves to the better neighbour,

28 $\rightarrow$ 22

③

step - 3 : check again

At the new solution, all neighbouring routes have
cost: ≥ 22 km

$\therefore$ no better neighbour exists.

Final cost = 22 km //

Search representation

Initial solution

SS

28 km

↓

Better Neighbour

22 km

↓

No better neighbour ✓

↓

Local optimum

It is not necessary that it is a global optimum.
Hill climbing checks only neighbouring solutions, so
22 km may be a local optimum.

Escaping Local optimum

Use random - restart hill climbing:

- start with a different route
- apply hill climbing again
- repeat & choose the best route ✓

Best solution found = global candidate //